A-A-51134A
25 November 1985
SUPERSEDING
A-A-51134
July 25, 1984

COMMERCIAL ITEM DESCRIPTION

BLADES, BAND SAW, COMPOSITE STEEL, METAL CUTTING

The General Services Administration has authorized the use of this commercial item description.

1. SCOPE

1.1 Scope. This description covers composite steel band saw blades of the standard, commercial type that are used for medium and heavy duty sawing of high alloy ferrous and non ferrous metals in a production environment.

1.2 Classification. Composite steel blades shall be of the following types, styles, and sizes listed in tables I through IX. The type, style, and size to be furnished shall be as specified (see 6.2).

Type I - M2 high speed steel

Style A - Standard tooth form (see table I).
Style B - Hook tooth form (see table II).
Style C - Variable tooth form (see table III).

Type II - M42 high speed steel

Style A - Standard tooth form (see table IV).
Style B - Hook tooth form (see table V).
Style C - Variable tooth form (see table VI).

Type III - Matrix high speed steel

Style A - Standard tooth form (see table VII).
Style B - Hook tooth form (see table VIII).
Style C - Variable tooth form (see table IX).

Size - sizes shall be selected from tables I through IX for band saw blades specified.

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TABLE I. Type I, style A, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.250	0.035	10	Raker
2	0.250	0.035	14	Raker
3	0.375	0.035	8	Raker
4	0.375	0.035	10	Raker
5	0.375	0.035	14	Raker
6	0.500	0.035	6	Raker
7	0.500	0.035	8	Raker
8	0.500	0.035	10	Raker
9	0.500	0.035	14	Raker
10	0.750	0.035	6	Raker
11	0.750	0.035	8	Raker
12	0.750	0.035	10	Raker
13	1.000	0.035	10	Wave
14	1.000	0.035	4	Raker
15	1.000	0.035	6	Raker
16	1.000	0.035	6	Wave
17	1.000	0.035	8	Raker
18	1.000	0.035	8	Wave
19	1.000	0.035	10	Raker
20	1.250	0.042	4	Raker
21	1.250	0.042	6	Raker
22	1.250	0.042	6	Wave
23	1.250	0.042	8	Raker
24	1.500	0.050	4	Raker
25	1.500	0.050	6	Raker

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TABLE II. Type I, style B, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.250	0.035	6	Standard
2	0.375	0.035	4	Standard
3	0.375	0.035	6	Standard
4	0.500	0.035	3	Standard
5	0.500	0.035	4	Standard
6	0.500	0.035	6	Standard
7	0.750	0.035	3	Standard
8	0.750	0.035	4	Standard
9	0.750	0.035	6	Standard
10	1.000	0.035	2	Standard
11	1.000	0.035	3	Standard
12	1.000	0.035	4	Standard
13	1.000	0.035	6	Standard
14	1.250	0.042	3	Standard
15	1.500	0.050	2	Standard
16	1.500	0.050	3	Standard
17	2.000	0.050	2	Standard
18	2.000	0.050	3	Standard

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TABLE III. Type I, style C, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.750	0.035	4-6	Standard
2	0.750	0.035	6-10	Standard
3	0.750	0.035	8-12	Standard
4	0.750	0.035	10-14	Standard
5	1.000	0.035	3-4	Standard
6	1.000	0.035	4-6	Standard
7	1.000	0.035	5-8	Standard
8	1.000	0.035	6-10	Standard
9	1.000	0.035	8-12	Standard
10	1.000	0.035	10-14	Standard
11	1.250	0.042	3-4	Standard
12	1.250	0.042	4-6	Standard
13	1.250	0.042	5-8	Standard
14	1.250	0.042	6-10	Standard

NOTE: Decimal pitch blades shall be rounded off to the nearest whole pitch blade.

TABLE IV. Type II, style A, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.750	0.035	4	Standard
2	0.750	0.035	6	Standard
3	0.750	0.035	8	Standard
4	0.750	0.035	10	Standard
5	1.000	0.035	3	Standard
6	1.000	0.035	4	Standard
7	1.000	0.035	6	Standard
8	1.000	0.035	8	Standard
9	1.250	0.042	3	Standard
10	1.250	0.042	4	Standard
11	1.250	0.042	6	Standard
12	1.250	0.042	8	Standard
13	1.500	0.050	4	Standard
14	1.500	0.050	6	Standard

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TABLE V. Type II, style B, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	1.000	0.035	3	Raker
2	1.000	0.035	4	Raker
3	1.000	0.035	6	Raker
4	1.250	0.042	3	Raker
5	1.250	0.042	4	Raker
6	1.500	0.050	2	Raker
7	1.500	0.050	3	Raker
8	2.000	0.050	2	Raker
9	2.000	0.050	3	Raker

TABLE VI. Type II, style C, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.750	0.035	6-10	Standard
2	1.000	0.035	3-4	Standard
3	1.000	0.035	4-6	Standard
4	1.000	0.035	6-10	Standard
5	1.250	0.042	2-3	Standard
6	1.250	0.042	3-4	Standard
7	1.500	0.050	2-3	Standard
8	1.500	0.050	3-4	Standard
9	1.500	0.050	4-6	Standard
10	2.000	0.050	2-3	Standard
11	2.000	0.050	3-4	Standard

NOTE: Decimal pitch blades shall be rounded off to the nearest whole pitch blade.

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TABLE VII. Type III, style A, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.250	0.035	10	Raker
2	0.375	0.035	10	Raker
3	0.500	0.035	10	Raker
4	0.500	0.035	14	Raker
5	0.750	0.035	6	Raker
6	0.750	0.035	8	Raker
7	0.750	0.035	10	Raker
8	0.750	0.035	10	Wave
9	1.000	0.035	4	Raker
10	1.000	0.035	6	Raker
11	1.000	0.035	6	Wave
12	1.000	0.035	8	Raker
13	1.000	0.035	8	Wave
14	1.000	0.035	10	Raker
15	1.000	0.035	10	Wave
16	1.000	0.035	14	Raker
17	1.250	0.042	4	Raker
18	1.250	0.042	6	Raker

TABLE VIII. Type III, style B, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.500	0.035	3	Raker
2	0.500	0.035	4	Raker
3	0.750	0.035	3	Raker
4	1.000	0.035	3	Raker
5	1.000	0.035	4	Raker
6	1.250	0.042	3	Raker
7	2.000	0.050	2	Raker
8	2.000	0.050	3	Raker

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TABLE IX. Type III, style C, sizes and characteristics.

Sizes	Characteristics			
	Width (inches)	Thickness (inches)	Pitch (teeth/inch)	Tooth set
1	0.500	0.035	5-8	Standard
2	0.500	0.035	6-10	Standard
3	0.500	0.035	8-12	Standard
4	0.500	0.035	10-14	Standard
5	0.750	0.035	3-4	Standard
6	0.750	0.035	4-6	Standard
7	0.750	0.035	5-8	Standard
8	0.750	0.035	6-10	Standard
9	0.750	0.035	8-12	Standard
10	0.075	0.035	10-14	Standard
11	1.000	0.042	2-3	Standard
12	1.000	0.042	3-4	Standard
13	1.000	0.042	4-6	Standard
14	1.000	0.042	6-10	Standard
15	1.250	0.042	2-3	Standard
16	1.250	0.042	3-4	Standard
17	1.250	0.042	4-6	Standard
18	1.250	0.042	5-8	Standard
19	1.250	0.042	6-10	Standard
20	1.250	0.042	8-12	Standard
21	1.500	0.050	2-3	Standard
22	1.500	0.050	3-4	Standard
23	1.500	0.050	4-6	Standard
24	1.500	0.050	5-8	Standard
25	1.500	0.050	6-10	Standard
26	2.000	0.050	3-4	Standard
27	2.000	0.050	4-6	Standard

NOTE: Decimal pitch blades shall be rounded off to the nearest whole pitch blade.

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2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals, form a part of this description to the extent specified herein. The documents should be obtained from the contracting activity or as directed by the contracting officer.

Federal Specification PPP-T-1150B	- Tools and Tool Accessories for Power Driven Metal and Woodworking Machinery, Packaging and Packing of.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.
American National Standards Institute, Standard ANSI B94.51	- Specifications for Band Saw Blades (Metal Cutting).
American Society for Testing and Materials Standard ASTM D 3951	- Standard Practice for Commercial Packaging.

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. SALIENT CHARACTERISTICS

3.1 Design. The band saw blades shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The band saw blades shall meet the requirements specified herein and shall conform to ANSI B94.51.

3.2 Material. The welded edge of the band saw blades of which the teeth are formed shall conform to the compositions listed in table X. The blade backing shall be manufactured of a low alloy spring steel that conforms to ANSI B94.51.

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TABLE X. Composition of band-saw blade edges.

Element	Type I		Type II		Type III	
	Min. %	Max. %	Min. %	Max. %	Min. %	Max. %
Carbon	0.78	0.88	1.05	1.15	0.70	0.75
Manganese	0.15	0.40	0.15	0.40	0.15	0.40
Phosphorus	----	0.03	----	0.03	----	0.03
Sulfur	----	0.03	----	0.03	----	0.03
Silicon	0.20	0.45	0.15	0.65	0.10	0.40
Chromium	3.75	4.50	3.50	4.25	3.90	4.40
Vanadium	1.75	2.20	0.95	1.35	0.80	1.10
Tungsten	5.50	6.75	1.15	1.85	0.80	1.20
Molybdenum	4.50	5.50	9.00	10.00	4.25	5.25
Cobalt	----	----	7.75	8.75	7.70	8.30

3.3 Measurement system. All dimensions are given in US units. These measurements may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

3.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered material to the maximum extent practical.

3.5 Dimensions and tolerances. Band saw blades shall conform to all dimensional and tolerance requirements contained in tables I through IX and ANSI B94.51. Tolerances not covered by ANSI B94.51 shall be width $\pm$ 0.005 inch and thickness $\pm$ 0.0005 inch.

3.6 Hardness. Blades shall have a body hardness of not less than Rockwell "C" 42 and not greater than Rockwell "C" 52. Type I blade tooth hardness shall be not less than Rockwell "C" 63 and not greater than Rockwell "C" 65. Type II blade tooth hardness shall be not less than Rockwell "C" 67 and not greater than Rockwell "C" 69. Type III blade tooth hardness shall be not less than Rockwell "C" 65 and not greater than Rockwell "C" 68.

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3.7 Length. Unless otherwise specified, band saw blades shall be furnished in coils of 100 feet. When specified (see 6.2), band saw blades shall be furnished in coil lengths of 250 feet, 500 feet, or in the welded length specified. When coils are supplied, there shall be no more than two welds each. When welded lengths are supplied there shall be no more than one weld each.

3.8 Performance. Each band saw blade shall be capable of meeting the sawing requirements specified in tables XI and XII. The test material shall be 1 x 3 inch flat bars of AISI 1018 cold drawn steel. The sawing test shall be performed with a blade tension of not less than 30,000 pounds per square inch, and not greater than 40,000 pounds per square inch. The test shall be performed on either a contour or a cut-off saw and conform to the following requirements.

3.8.1 Contour saw. The test blade shall be mounted on a contour saw of not less than a two horsepower rating and a full-flow or mist-type fluid delivery system. With a blade speed of 250 feet per minute, twenty cuts shall be made, the first five of which shall be made within the time limits specified in table XI in the column with the heading "Break in" time. During the next few cuts the feed pressure shall be gradually increased so that on the tenth cut the time shall be within the limits specified in table XI in the column with the heading "Test". Without further adjustment to the saw, ten cuts shall be made all of which shall be within the time limits specified in table XI in the column with the heading "Test".

3.8.2 Cut-off saw. The test blade shall be mounted on a cut-off band saw machine of not less than a five horsepower rating and a water soluble "HO" type fluid delivery system. With a blade speed of 250 feet per minute, thirty cuts shall be made, the first five of which shall be made within the time limits specified in table XII in the column with the heading "Break in" time. During the next few cuts the feed pressure shall be gradually increased so that on the tenth cut the time shall be within the limits specified in table XII in the column with the heading "Test". Without further adjustment to the saw, twenty cuts shall be made all of which shall be within the time limits specified in table XII in the column with the heading "Test".

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TABLE XI. Band saw blade test requirements, contour saw.

Width	Pitch (number of teeth per inch)		Workpiece position	Time of cut (in seconds)	
	Standard or hook tooth	Variable pitch		Break in	Test
0.250/0.500	3		Upright	90-100	60-70
0.250/0.500	4		Flat	100-120	70-80
0.250/0.500	6	4-6	Flat	100-120	70-80
0.250/0.500	8	5-8	Flat	120-150	80-90
0.250/0.500	10		Flat	120-150	80-90
0.250/0.500	14		Flat	120-150	80-90

NOTE: When in the upright position the workpiece shall be resting on the one inch dimensional edge, when in the flat position the workpiece shall be resting on the three inch dimensional side.

TABLE XII. Band saw blade test requirements, cut-off saw.

Pitch (number of teeth per inch)		Number of workpieces	Time of cut (in seconds)	
Standard or hook	Variable pitch		Break in	Test
2		5	100-180	50-55
3	2-3	4	80-150	40-45
4	3-4	3	60-100	30-35
6	4-6	2	50-90	25-30
	5-8	2	50-90	25-30
8	6-10	1	30-60	15-20
10	8-12	1	30-60	15-20
	10-14	1	50-90	25-30
14		1	50-90	25-30

NOTE: Workpieces shall be positioned side by side resting on the one inch dimensional edge.

3.9 Workmanship. Workmanship of the band saw blades shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the description where such inspections are deemed necessary to assure supplies.

4.2 Sample inspection. The supplier shall randomly select one sample blade from each lot of blades offered for acceptance to determine conformance with the examination of 4.3, the dimensional inspection in 4.4, and the hardness test in 4.5. When specified (see 6.2), the sample blade shall also be subjected to the performance test in 4.6. If the sample blade passes all examination, inspection, and test requirements specified, the entire lot shall be accepted and considered suitable for shipment. If the sample blade fails to pass all examination, inspection, and test requirements specified, the supplier shall select two additional sample blades from the same lot. The two additional sample blades shall be subjected to the same examination, inspection, and test requirements that the failed blade was subjected to. If both additional sample blades pass all examination, inspection, and test requirements specified, the supplier shall replace the defective blade with a blade that meets all the requirements specified herein and the entire lot shall be accepted and considered suitable for shipment. If either of the two additional sample blades fails to pass all examination, inspection and test requirements specified, the entire lot shall be rejected and considered unsuitable for shipment. A lot is defined as all blades of the same type, style, and size presented at one time for acceptance.

4.3 Examination. Each band saw blade submitted for sample inspections shall be visually examined to determine conformance with all requirements in this description.

4.4 Dimensional inspection Each sample band saw blade shall be inspected to assure conformance with all dimensional and tolerance requirements specified in tables I through IX and ANSI 894.51 (see 3.5).

4.5 Hardness test. Each sample band-saw blade shall be subjected to a hardness test to determine conformance with the applicable requirements in 3.6. If the hardness test is destructive the blade used for the hardness test shall not be considered as part of the procurement order.

4.6 Performance test. The performance test shall conform to the requirements of 3.8. The cutting force and machining techniques shall be in accordance with standard industrial procedures. After performing the test the band saw blade shall show no evidence of overheating or physical damage. The band saw blades used for the performance test shall not be counted as part of the procurement order.

4.7 Packaging inspection. Packaging shall be inspected for conformance with section 5.

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5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), preservation, packing, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

6. NOTES

6.1 Intended use. Band saw blades covered by this description are intended to be used for sawing ferrous and non-ferrous metals.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this description.

b. Type, style, size, and quantity required, (see 1.2).

c. Length, if different (see 3.7).

d. Performance test, if required (see 4.2).

e. Packaging in accordance with PPP-T-1150 if required, specify level (see 5.1).

MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review Activities

Army - AL
Navy - YD
Air Force - 84
DLA - GS

User Activities

Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT - NHT
GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3455-0100

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Memphis, TN 38114-5297

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51134A	Blades, Band Saw, Composite Steel, Metal Cutting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Defense Industrial Plant Equipment Center	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)* Airways Blvd. Memphis, TN 38114-5297	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
DIPEC-SSG Defense Industrial Plant Equip. Cntr	
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional Airways Blvd. Memphis, TN 38114-5297	8. DATE OF SUBMISSION *(YYMMDD)* 27 Nov 85

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